DemoRecruit: Budgeted Participant Recruitment for Crowdsensing

I. PARTICIPANT SELECTION

This is an original synthetic document for software demonstrations. DemoRecruit and ToyRoutes are fictional. The procedure below is illustrative, not a published algorithmic contribution.

DemoRecruit selects crowdsensing participants under a fixed budget. Each candidate provides an expected route and a requested payment. At each step, the selector computes newly covered cells divided by payment and selects the highest ratio among affordable candidates.

After each selection the covered-cell set is updated. Participants whose payments exceed the remaining budget are skipped. Selection stops when no affordable participant contributes a new cell. The example does not claim global optimality.

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II. DATA AND FAILURE CASES

ToyRoutes is a fictional dataset containing six candidate routes over a small grid. The evaluation compares covered cells with the requested payments, rather than measuring sensor-reading reconstruction error.

If every route covers the same cells, selecting additional participants may add no coverage. The procedure then stops even if budget remains. Route uncertainty and malicious location reports are limitations of this simplified example.

This document does not contain a trained reinforcement learning policy, real user identifiers or quantitative deployment results.